Algorithms for Big Data

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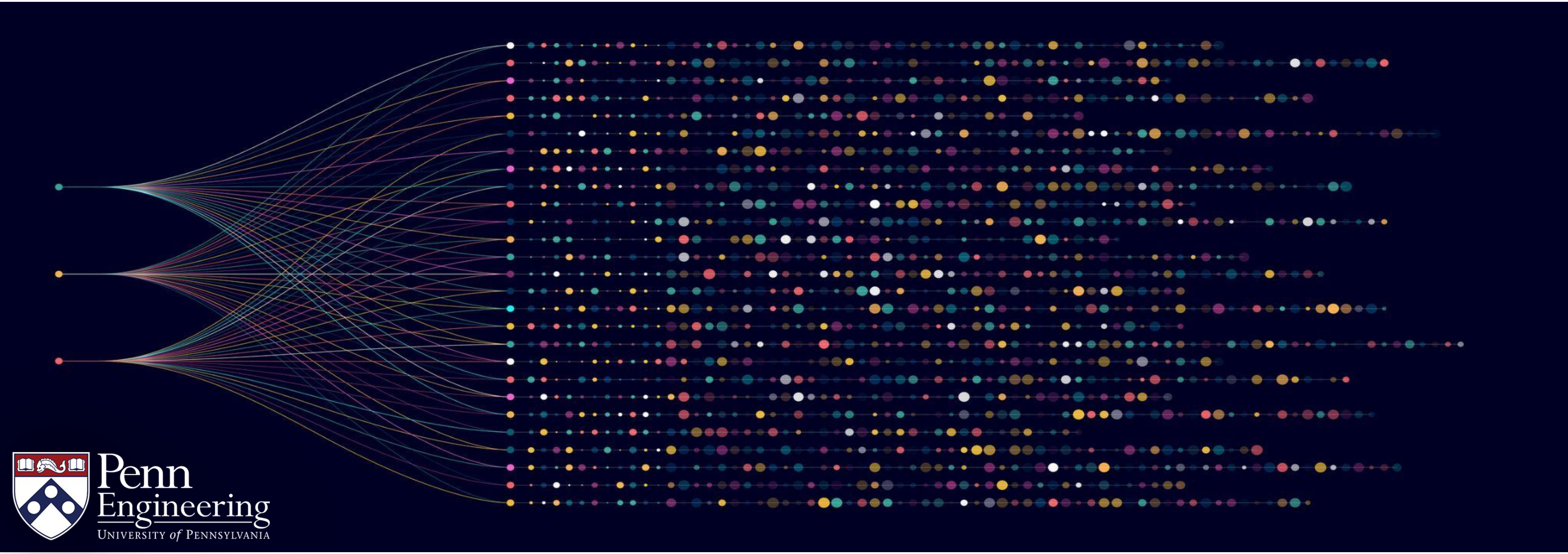
Probabilistic Guarantees



Objectives: Computing Large Averages

- Probabilistic guarantees for algorithms

- Additive approximations with high probability

Guarantees of Randomized Algorithms

Definition: A randomized algorithm for approximating an average is an algorithm with the following guarantees:

- Input:
  - Query access to $X_1, \ldots, X_n \in \mathbb{R}$
  - Accuracy parameter ϵ
  - Source of randomness
  - Failure probability $\delta \in (0, 1)$
- Guarantee: $\Pr\left[\left|\hat{\boldsymbol{S}} - \frac{1}{n}\sum_{i=1}^{n} X_i\right| \leq \epsilon\right] \geq 1 - \delta.$

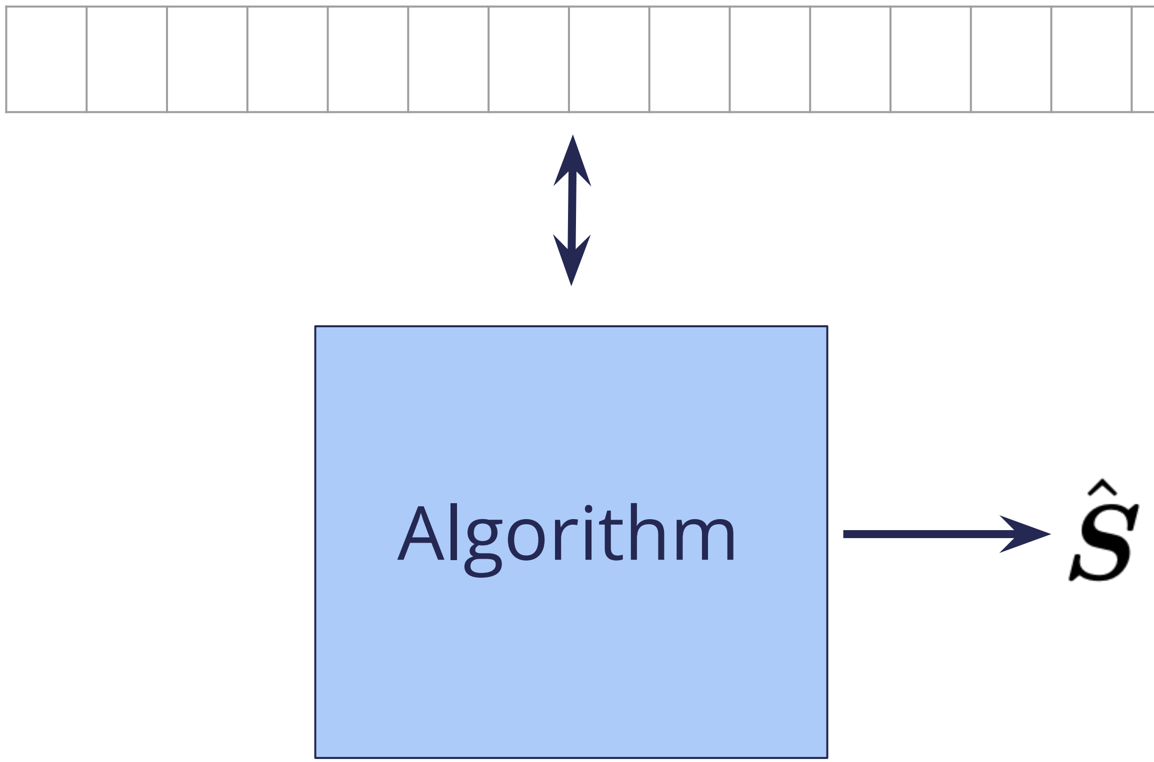
Interpreting results of randomized approximations

- How many users are in Europe?
    - Suppose accuracy is 0.05 and failure probability is 0.01
    - Output of algorithm is 0.3
    - There is a 1% chance algorithm is incorrect. If correct, then between 25% and 35% of users are in Europe.

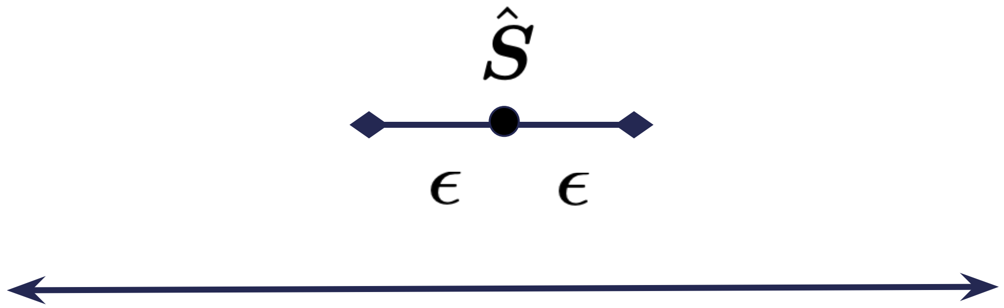
- How many users posted in the past week?
    - Suppose accuracy is 0.1 and failure probability is 0.05
    - Output of algorithm is 0.9
    - There is a 5% chance algorithm is incorrect. If correct, then between 80% and 100% of users posted in the past week.

- How long (on average) did users spend on the platform today?
    - Suppose accuracy is 30 and failure probability 0.0001
    - Output of algorithm is 120
    - There is a 0.01% chance algorithm is incorrect. If correct, then users spent between 90 and 150 seconds on the platform on average.

Approximate and randomized guarantees: **user perspective**



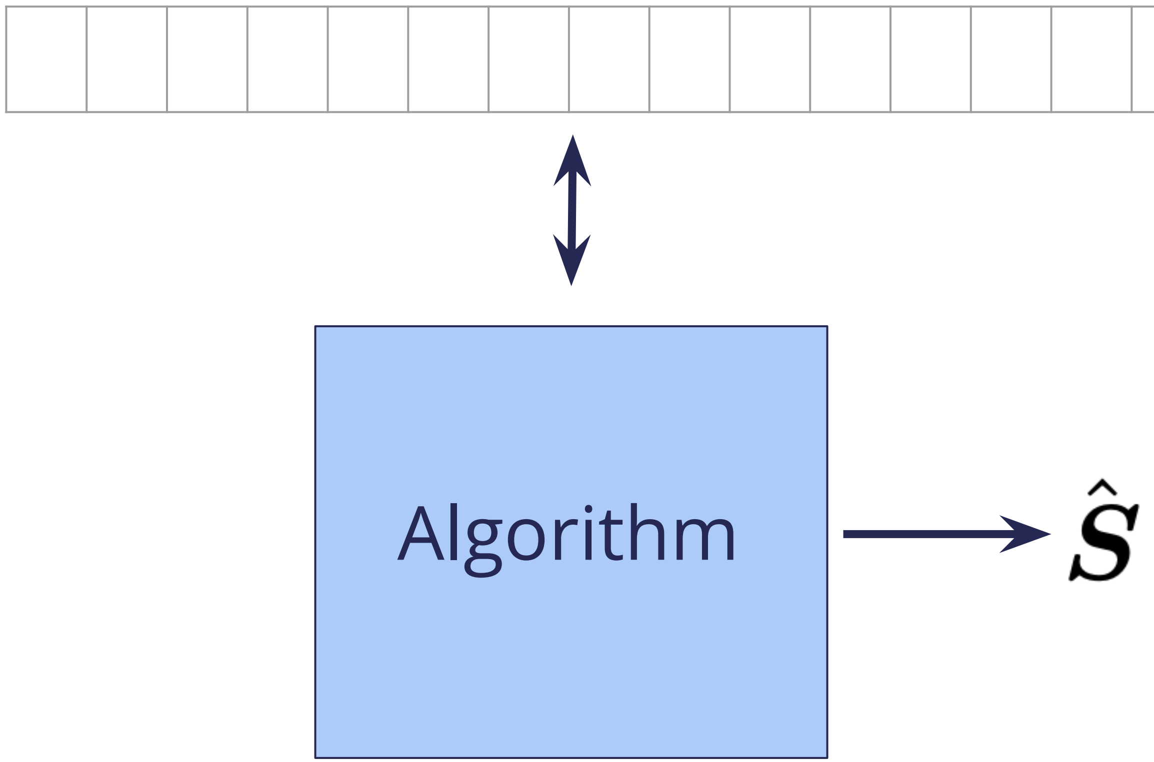
may be wrong with probability δ

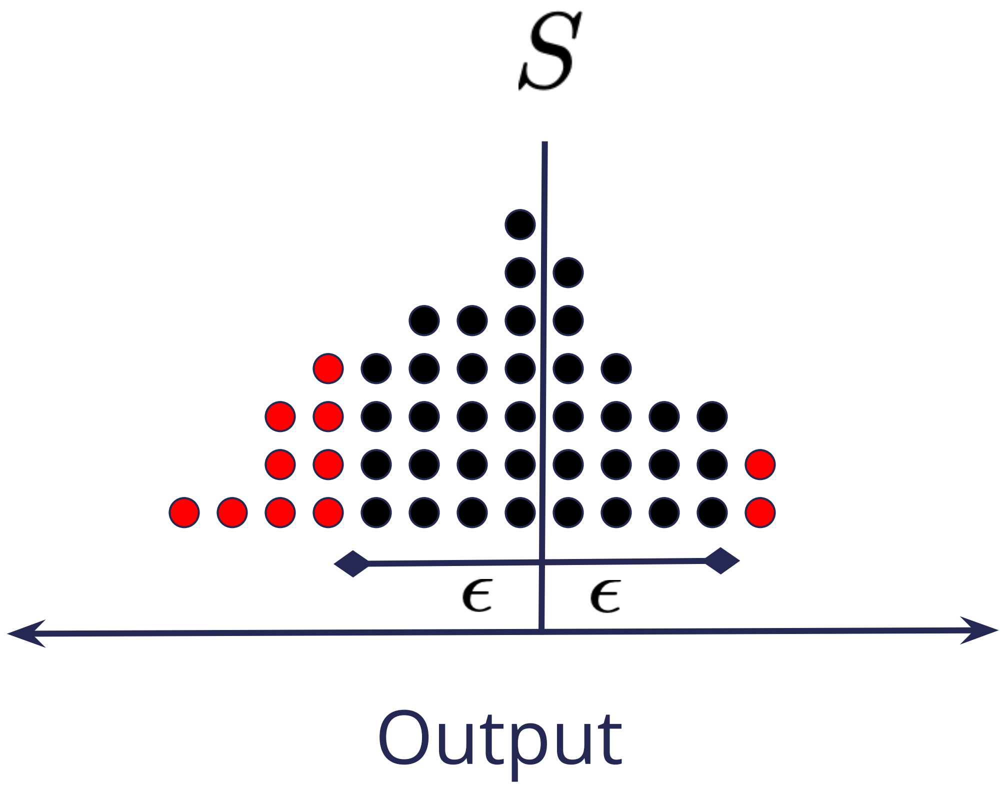


Output

ϵ : accuracy δ: failure probability

Approximate and randomized guarantees: **algorithm perspective**





ϵ : accuracy δ: failure probability

Key algorithmic questions:

- Do there exist (more) efficient algorithms when allowing approximation and randomization?
- What is the tradeoff between the number of queries, the accuracy parameter, and failure probability?